T20 DATA USAGE vs IPL DATA USAGE

Problem with each Approach:

IPL DATA USAGE: The fundamental issue is the lack of direct IPL data for players who haven't played IPL yet. Therefore, neither can we train the model for them neither can we predict fantasy points for them

T20 DATA USAGE: The intensity, player quality, and playing conditions (pitches, grounds) differ significantly between IPL and other leagues like SMA, BBL, CPL, etc. A high strike rate in SMA might not translate directly to the same level of explosiveness in the IPL against higher quality bowling attacks.

Possible Solutions:

1. League-Specific Normalization/Scaling:

- **Idea:** For each feature, calculate league-specific benchmarks or distributions. Then, scale or normalize a player's performance in a non-IPL league relative to these benchmarks to estimate their potential IPL performance.
- **Steps:**
    1. **Calculate League Statistics:** For each league in your dataset (IPL, SMA, BBL, CPL, SA20, Hundred), calculate the average (or median, or even percentiles) for each of your features (Runs per match, Average, Strike Rate, Boundaries per match). You might want to segment this by batting position (e.g., top-order batsman, middle-order, finisher) for more granular insights.
        - **Normalization/Scaling:** For a player with no IPL data, calculate the ratio of their feature value in their respective league to the average feature value in that league. Then, multiply this ratio by the average feature value in the IPL for players in a similar role/batting position.
            - Example (Strike Rate):
                - Average SR in SMA = 120
                - Average SR in IPL = 135

        - Indian player's SR in SMA = 140
        - Estimated IPL SR = (135 / 120) * 140 = 157.5

    2. **Training:** Train your prediction model using primarily IPL data for players who have played in the IPL. For players with no IPL data in your training set (this might be tricky), you could potentially use their normalized/scaled values as if they were IPL data points, but be cautious about the reliability.

- **Pros:** Relatively simple to implement and understand. Directly addresses the difference in league quality.
- **Cons:** Assumes a linear relationship between league performance. Might not capture the nuances of transitioning to the IPL (pressure, different bowling styles, etc.). Choosing the right benchmark (average, median, etc.) and the scaling method (ratio vs. difference) requires experimentation.

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2. Two-Stage Model:

This approach involves building two separate models.

- **Idea:** First, build a model to predict a player's "underlying skill" based on all available T20 data. Then, build a second model to predict IPL performance based on this "underlying skill" and potentially other IPL-specific factors.
- **Steps:**

    1. **Skill Prediction Model:** Train a model using data from all leagues (IPL, SMA, BBL, etc.) to predict a latent "skill score" for each player. The target variable here could be a composite metric derived from your features or even something like an Elo rating adapted for T20.

    2. **IPL Performance Model:** Train a second model using only IPL data. The input features for this model would include the "skill score" predicted in the first stage, along with other potentially relevant IPL-specific features (e.g., team composition, venue, opponent).

- **Pros:** Allows for a more nuanced understanding of a player's overall T20 ability before predicting their IPL performance. Can potentially incorporate more complex relationships.

- **Cons:** More complex to implement. Defining and training the "skill score" can be challenging. The accuracy of the second model heavily depends on the quality of the first model's predictions.

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3. Feature Engineering with League Indicators:

- **Idea:** Include the league information as a feature in your training data. This allows the model to learn league-specific effects for each player.
- **Steps:**
    1. **Add League Feature:** Create a categorical feature indicating the league from which each data point originates (e.g., "IPL", "SMA", "BBL").
    2. **Train on All Data:** Train your prediction model using all available T20 data (IPL and other leagues), including the league feature.
    3. **Prediction:** For players with no IPL data, their data points will have the corresponding league (e.g., "SMA" or "BBL") in the league feature. The model should ideally learn to adjust the predictions based on the league.
- **Pros:** Relatively straightforward to implement. Allows the model to implicitly learn the differences between leagues.
- **Cons:** Might require a large amount of data from various leagues to learn the nuances effectively. The model might struggle to generalize to players from leagues with very limited data.

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4. Transfer Learning (More Advanced):

- **Idea:** Train a model on a large dataset of all T20 data and then fine-tune it specifically on IPL data. This can help leverage the general T20 knowledge learned from the larger dataset while specializing for the IPL.
- **Steps:**
    1. **Pre-training:** Train a base model on all available T20 data (IPL and other leagues).
    2. **Fine-tuning:** Take the pre-trained model and further train it using only IPL data. You might adjust the learning rate or freeze some layers during fine-tuning.

3. **Prediction:** Use the fine-tuned model for predictions on IPL matches. For players without IPL data, their non-IPL data would have been part of the pre-training phase, and the model would have learned some representation for them.

- **Pros:** Can potentially lead to better performance by leveraging a larger dataset.
- **Cons:** More complex to implement and requires a good understanding of transfer learning techniques.

Predicting for a Match After Training:

Once you have trained your model using one of the above approaches (or a combination), here's how you would predict the target variable :

1. **Identify Playing XI:** For the upcoming IPL match, determine the playing XI for each team along with details such as player roles and credit.

2. **Feature Extraction:** For each player in the probable XI:
    - **If the player has IPL history:** Extract their relevant historical IPL data to calculate your chosen features based on their past IPL performances.
    - **If the player has NO IPL history:**
        - **Indian Player:** Use their recent SMA data to calculate the features.
        - **International Player:** Use their recent data from relevant international T20 leagues (BBL, CPL, SA20, Hundred, etc.). You might need to decide how to aggregate data if a player has played in multiple leagues recently (e.g., average the features).

3. **Data Preprocessing:** Apply any necessary preprocessing steps to your extracted features, such as the normalization or scaling you used during training.

4. **Model Prediction:** Feed the pre-processed features into your trained model to get a prediction for each player. The output of the model could be a predicted number fantasy points or whether a player is in Dream Team or not, depending on your model's design.